University of Science

Faculty of Information Technology

# Data Structures and Algorithms Project Report

**Topic: STRING MATCHING**

**Course: Data Structures & Algorithms**

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# 1 Group Information

| **Student Name** | **ID** | **Assigned Tasks** | **% Completed** |
|---|---|---|---|
| Dang Ba Lam | 25127403 | Leader, Command line arguments, Implementation & Report (The Naive approach, Experimental results) | 25% |
| Hua Minh Khang | 25127073 | Setting Algorithm mode and Comparisons mode, Make input for Input Size, Implementation & Report (The Knuth-Morris-Pratt algorithm, The Aho Corasick algorithm Acknowledgement) | 25% |
| Trinh Bao Khoi | 25127256 | Data Generator, Implementation & Report (The Boyer-Moore algorithm, Project organization) | 25% |
| Nguyen Huu Ngoc Thuan | 25127153 | Adjust Comparisons and Time, Implementation & Report (The Rabin-Karp algorithm, Title Page, Table of Contents, Team Members List) | 25% |
| | | **Total Group Completion** | **110%** |

Table 1: Group Information

**Note: All members participated in Time Calculator implementation, Report writing, Citations, and Final checking.*

# 2 Problem Statement

STRING matching is a problem that find all occurrences of a pattern in the text. This problem is an indispensable technique in computer science for solving various crucial problems, such as text editing, text searching, and symbol manipulation.[3]

The string matching problem is defined as follows: We assume that the text is a string $T[1..n]$ of length $n$ and the pattern is a string $P[1..m]$ of length $m$ which $m \leq n$. The problem is to find all occurrences of $P$ in $T$ by shifting $P$ along $T$ such that $0 \leq s \leq n - m$ ($s$ is the shift amount) and $T[s+1..s+m] = P[1..m]$ (that is, if $T[s+i] = P[i]$, for $1 \leq i \leq m$).[2]

For instance:

- If $T =$ "ababcab" and $P =$ "abc", then the output is $(3 \rightarrow 5)$ because the pattern occurs at position 3 in the text.

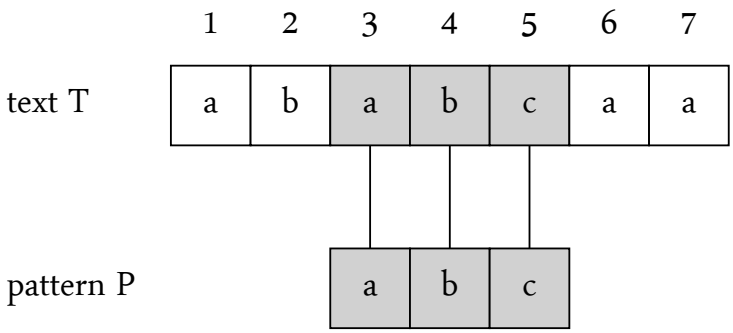



Figure 1: T = "ababcab" and P = "abc"

- If $T =$ "aaaaa" and $P =$ "aa", then the output is $(1 \rightarrow 2)$, $(2 \rightarrow 3)$, $(3 \rightarrow 4)$, and $(4 \rightarrow 5)$ because the pattern occurs at positions 1, 2, 3, and 4 in the text.

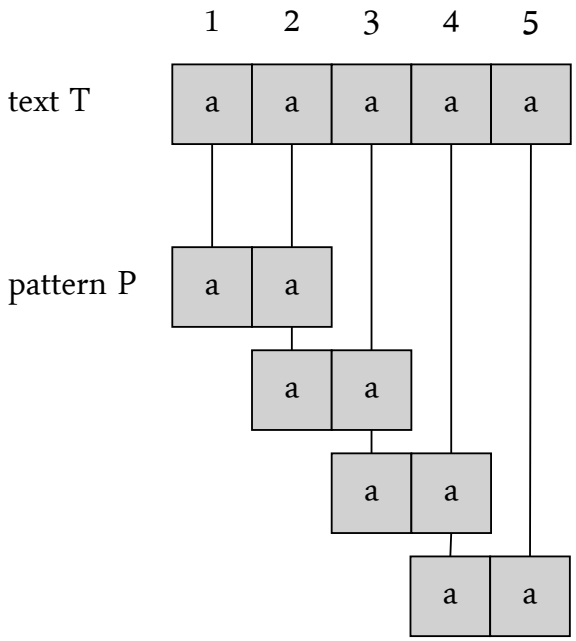



Figure 2: T = "aaaaa" and P = "aa"

# 3 Algorithms Analysis

## 3.1 The Naïve string-matching algorithm (Brute-Force approach)

### 3.1.1 Introduction & Application

- **Core idea:** According to Abdeen[1], the brute-force algorithm works with two pointers; a "text pointer" i and a "pattern pointer" j. For all possible shifts of the pattern P over the text T, compare the pattern characters with the corresponding text characters. If a mismatch occurs, i is incremented by 1, and j is reset to 0 and the comparing process is restarted. In the case that all characters match, the algorithm records the position of the match and continues searching for the next possible match.

- **Real-world applications:**
  - Text Editors: There are used in "Search" or "Find and Replace" features to locate target words.
  - Bioinformatics: Searching for specific DNA sequences.
  - Checking for Plagiarism: Comparing documents to find and compare a script or a piece of code with the exact match in a large database.

### 3.1.2 Pseudocode

**Notation:**

- **Text $T$ (R rows, C columns)**: The main text being searched.
- **Pattern $P$ (K: the number of keywords)**: The list of strings that we are looking for.
- **Positions ([(start position, end position), ...])**: A list storing the indices where a complete match is found.

```
BRUTE-FORCE-ALGORITHM(T, P, K, R, C)
pos = empty array of size K
for idx = 1 to K do
    keyword = P[idx]
    m = keyword.length

    ▷ Search Horizontally
    for i = 1 to R do
        for j = 0 to C − m do
            if keyword[1 ... m] == T[i, j + 1 ... j + m] then
                add position to pos[idx]
            end if
        end for
    end for

    ▷ Search Vertically
    for i = 1 to C do
        for j = 0 to R − m do
            if keyword[1 ... m] == T[j + 1 ... j + m, i] then
                add position to pos[idx]
```

```
            end if
        end for
    end for
end for
return pos
```

### 3.1.3 Step-by-step trace[2]

**Notation:**

- **Text** $T$ **(length** $n$**)**: The main text being searched.
- **Pattern** $P$ **(length** $m$**)**: The string that we are looking for.
- **Shift** ($s$): The current offset of the text T.
- **Positions ([(start position, end position), . . . ])**: A list storing the indices where a complete match is found.

**Example:** text $T = [a, a, b, c, a, b, b]$ and pattern $P = [a, b, c]$

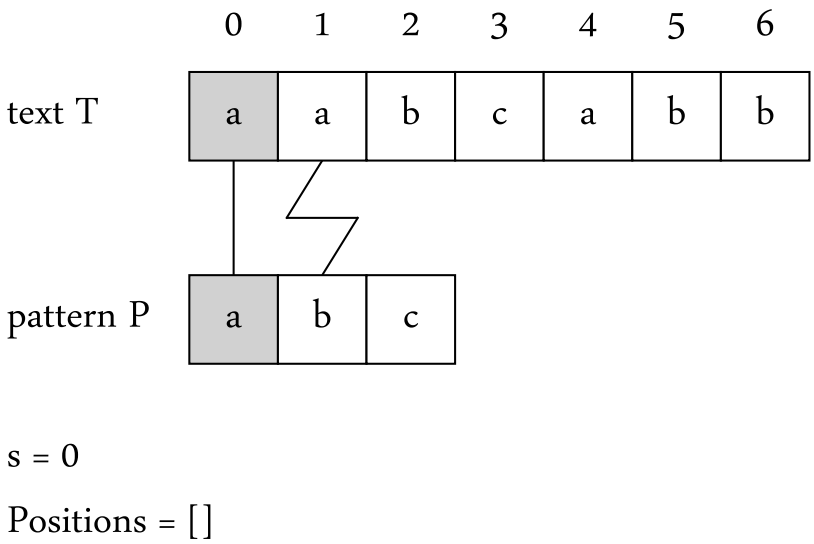
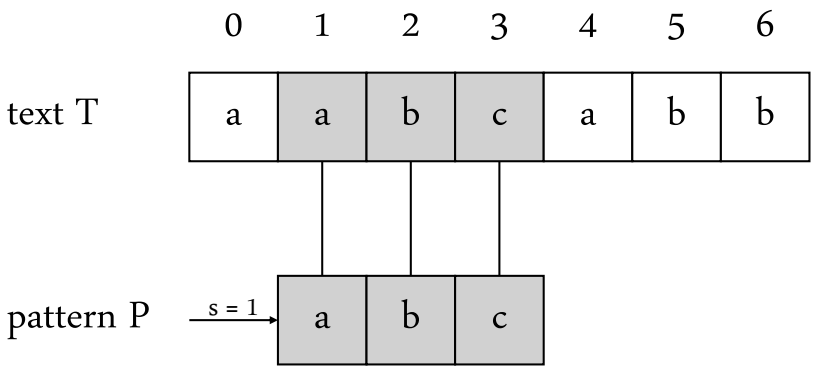



Figure 3: Step 1

- The first character 'a' matches between the pattern and the text.

  The second character in text $T$ ('a') does not match the second character in pattern $P$ ('b').

  Text at the bottom shows the current shift is $s = 0$ and the list of found matches is empty ($Positions = []$).



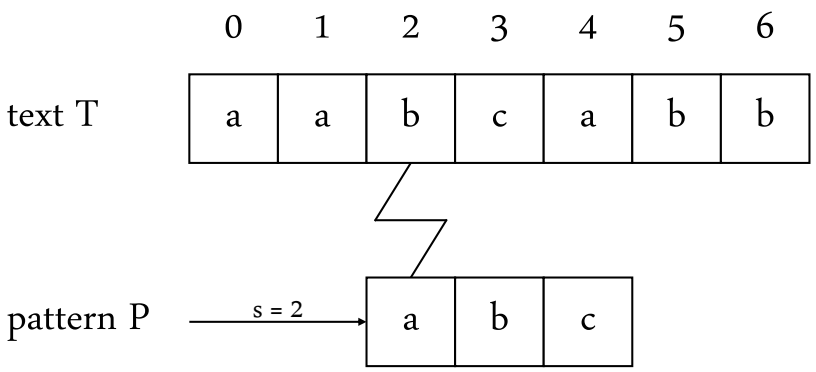

s = 1

Positions = [(1, 3)]

Figure 4: Step 2

- All characters in pattern $P$ ('a', 'b', 'c') successfully match the corresponding characters in text $T$ ('a', 'b', 'c').

  The shift is $s = 1$ and the list of found matches is updated to record this position ($Positions = [(1, 3)]$).



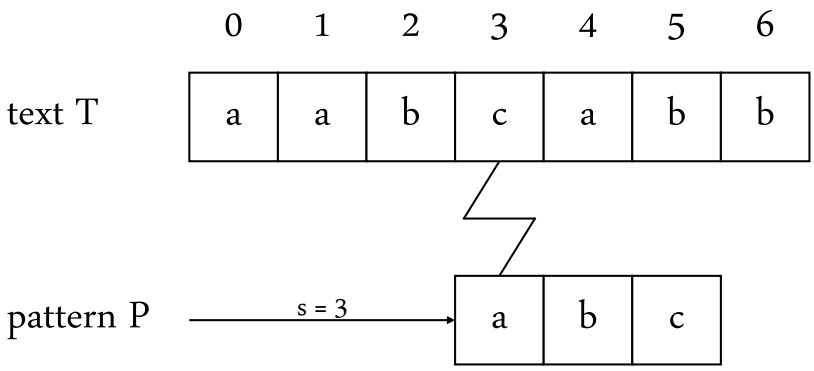

s = 2

Positions = [(1, 3)]

Figure 5: Step 3

- The first character in pattern $P$ ('a') does not match the character in text $T$ ('b') at the current shift position $s = 2$.

  Skip to the next shift without updating $Positions$.



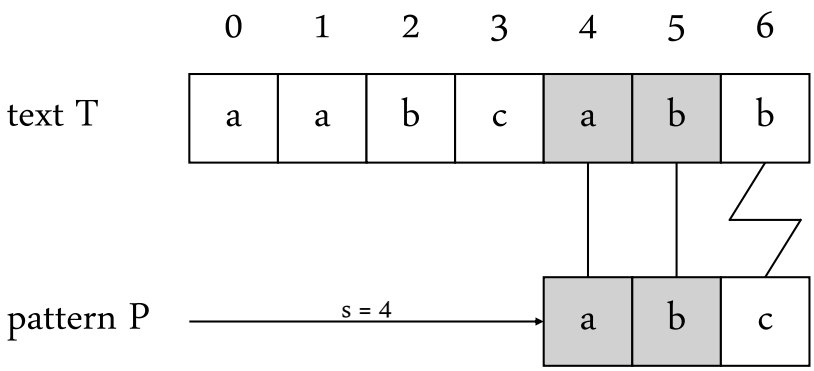

s = 3

Positions = [(1, 3)]

Figure 6: Step 4

- The first character in pattern $P$ ('a') does not match the character in text $T$ ('c') at the current shift position $s = 2$.

  Skip to the next shift without updating *Positions*.




s = 4

Positions = [(1, 3)]

Figure 7: Step 5

- The first two characters in pattern $P$ ('a', 'b') match the corresponding characters in text $T$, but the third character in pattern $P$ ('c') does not match the character in text $T$ ('b') at the current shift position $s = 4$.

  Skip to the next shift without updating *Positions*.

- After that, $s$ increases by 1. Because $s = 5$ and $n - m = 7 - 3 = 4$, we have $s > n - m$. The algorithm terminates and returns the list of found matches: Positions $= [(1, 3)]$.

### 3.1.4 Complexity

| Metric | Complexity | Reasoning |
|---|---|---|
| **Time (Average)** | $O(K \cdot R \cdot C)$ | In practice, a mismatch usually occurs after only a few character comparisons. |
| **Time (Best with flag optimization)** | $O(K \cdot R \cdot C)$ | The first character of the pattern always results in a mismatch at every shift. |
| **Auxiliary Space** | $O(1)$ | The algorithm uses only a constant amount of extra memory for loop indices. |

Table 2: Complexity Analysis of 2D Brute Force Algorithm

### 3.1.5 Discussion

- In the basic brute-force implementation, the time complexity remains consistent across different scenarios.
- However, there is a way to optimize the brute-force algorithm by skipping some shifts when a mismatch occurs. This can be achieved by using a flag.
  - In Scenario 1, the number of patterns $K$ is small, and the text is from $10 \times 10$ to $500 \times 500$. Brute-force approach is efficient and likely linear running time $O(R \cdot C)$.
  - In Scenario 2, the number of patterns $K$ is large, and the text is fixed at $100 \times 100$. This scenario is more challenging for the brute-force approach, as the running time multiplies by $K$ times, which is much longer compared to scenario 1.
  - In conclusion, the brute-force approach is more suitable for scenarios with a small number of patterns. It is worse for scenarios with a large number of patterns.

## 3.2 The Knuth-Morris-Pratt algorithm (KMP algorithm)

### 3.2.1 Introduction & Application

- **Core idea:** According to Abdeen[1], the brute-force algorithm works with two pointers; a "text pointer" i and a "pattern pointer" j. For all possible shifts of the pattern P over the text T, compare the pattern characters with the corresponding text characters. If mismatch occurs, i is incremented by 1, and j is reset to 0 and the comparing process is restarted. In the case that all characters match, the algorithm records the position of the match and continues searching for the next possible match.
- **Real-world applications:**
  - Text Editors: Using in "Search" or "Find and Replace" features to locate destination words.
  - Bioinformatics: Searching for specific DNA.
  - Checking for Plagiarism: Comparing documents to find and compare a script or a piece of code with the exact match in a large database.

### 3.2.2 Pseudocode

**Notation:**

- **Text $T$ (R rows, C columns)**: The 2D grid of characters.
- **Pattern $P$ (K keywords)**: The list of patterns to search for.
- **Prefix Array $\pi$**: An array storing the lengths of the longest proper prefix that is also a suffix.

```
CONVERT-GRID-TO-STRINGS(T, R, C)
ans = empty array
    ▷ Extract Rows
for i = 0 to R − 1 do
    tmpString = empty string
    for j = 0 to C − 1 do
        tmpString = tmpString + T[i][j]
    end for
    append tmpString to ans
end for
    ▷ Extract Columns
for j = 0 to C − 1 do
    tmpString = empty string
    for i = 0 to R − 1 do
        tmpString = tmpString + T[i][j]
    end for
    append tmpString to ans
end for
return ans
```

```
CALCULATE-POSITION(idxString, m, R, idx)
if idxString < R then
    ▷ Horizontal match (Row extraction)
    start = (idxString, idx − 2 · m)
    end = (idxString, idx − m − 1)
else
    ▷ Vertical match (Column extraction)
    start = (idx − 2 · m, idxString − R)
    end = (idx − m − 1, idxString − R)
end if
return (start, end)
```

```
KMP-ALGORITHM(T, P, K, R, C)
pos = array of K empty lists
List_string = CONVERT-GRID-TO-STRINGS(T, R, C)
N = List_string.length

for idxString = 0 to N − 1 do
    for pIdx = 0 to K − 1 do
        kmpString = P[pIdx] + '#' + List_string[idxString]
        L = kmpString.length
        m = P[pIdx].length
```

```
        π = array of size L initialized to 0
    ▷ Compute Prefix Function & Search
        for i = 1 to L − 1 do
            j = π[i − 1]
            while j > 0 and kmpString[j] ≠ kmpString[i] do
                j = π[j − 1]
            end while
            if kmpString[j] == kmpString[i] then
                j = j + 1
            end if
            π[i] = j
    ▷ Match found
            if π[i] == m then
                matchPos = CALCULATE-POSITION(idxString, m, R, i)
                append matchPos to pos[pIdx]
            end if
        end for
    end for
end for
return pos
```

### 3.2.3 Step-by-step trace[2]

**Notation:**

- **Text** $T$ **(length** $n$**)**: The main text being searched.
- **Pattern** $P$: The string that we are looking for.
- **Shift** ($s$): The current offset of the text T.
- **Positions ([(start position, end position), . . . ])**: A list storing the indices where a complete match is found.

**Example:** text $T = [a, a, b, c, a, b, b]$ and pattern $P = [a, b, c]$

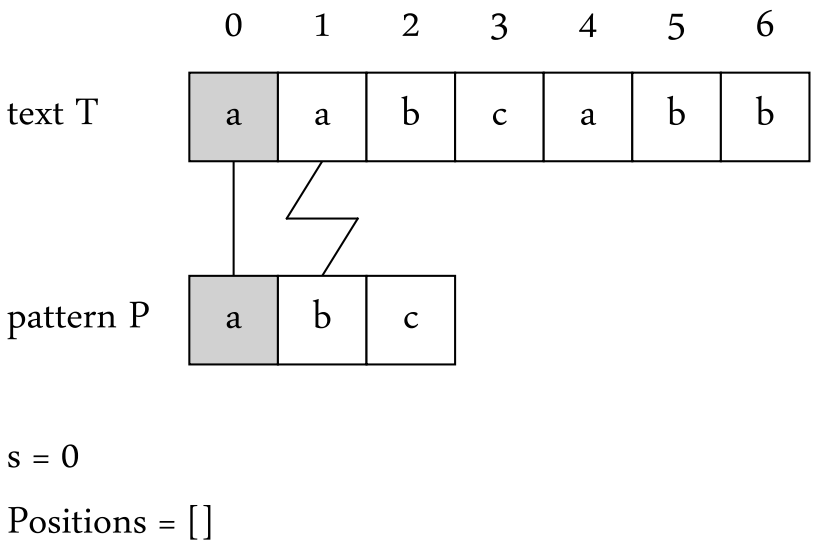



Figure 8: Step 1

- The first character 'a' matches in both arrays.

  The second character in text $T$ ('a') does not match the second character in pattern $P$ ('b').

  Text at the bottom shows the current shift is $s = 0$ and the list of found matches is empty ($Positions = []$).

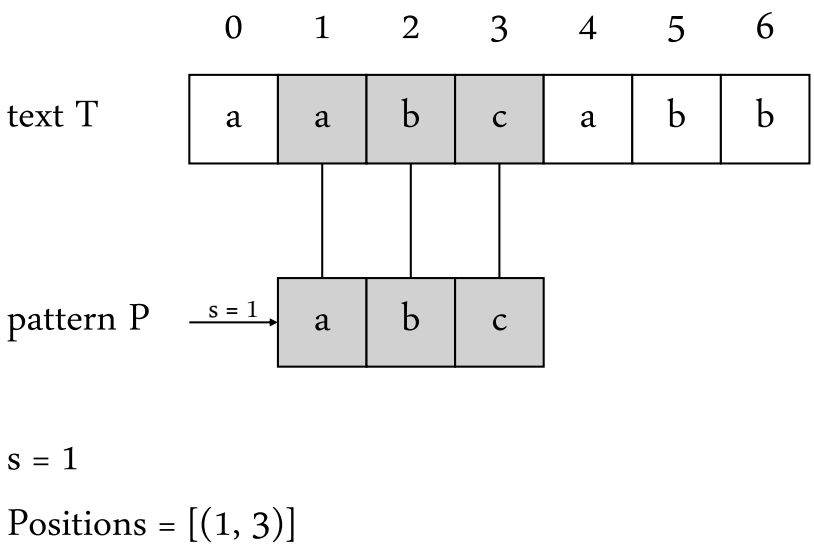
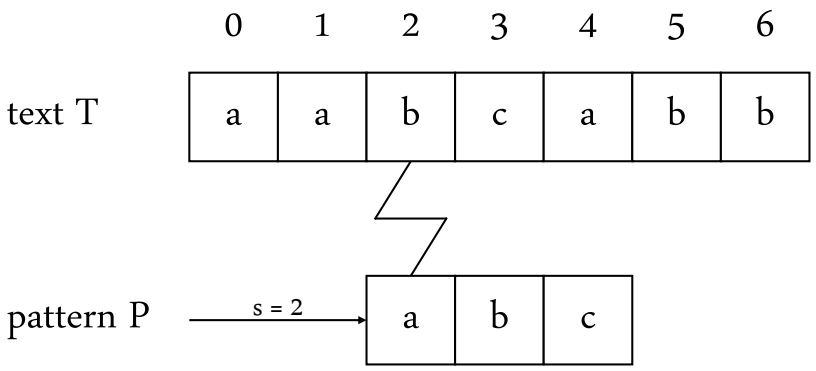



Figure 9: Step 2

- All characters in pattern $P$ ('a', 'b', 'c') successfully match the corresponding characters in text $T$ ('a', 'b', 'c').

  Shift is $s = 1$ and the list of found matches is updated to record this position ($Positions = [(1, 3)]$).



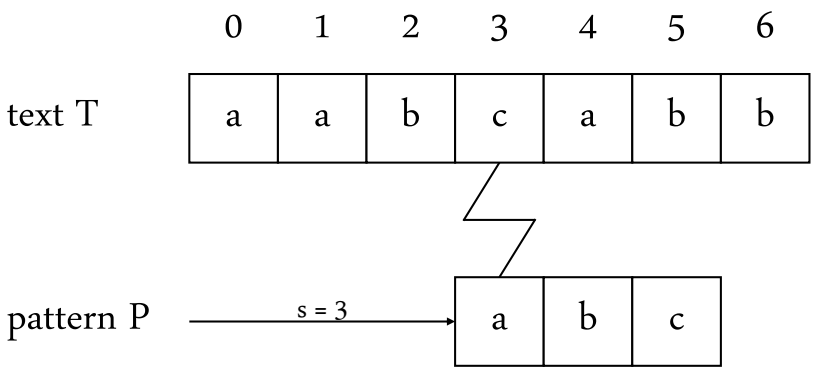

s = 2

Positions = [(1, 3)]

Figure 10: Step 3

- The first character in pattern $P$ ('a') does not match the character in text $T$ ('b') at the current shift position $s = 2$.

  Skip to the next shift without updating *Positions*.




s = 3

Positions = [(1, 3)]

Figure 11: Step 4

- The first character in pattern $P$ ('a') does not match the character in text $T$ ('c') at the current shift position $s = 2$.

  Skip to the next shift without updating *Positions*.

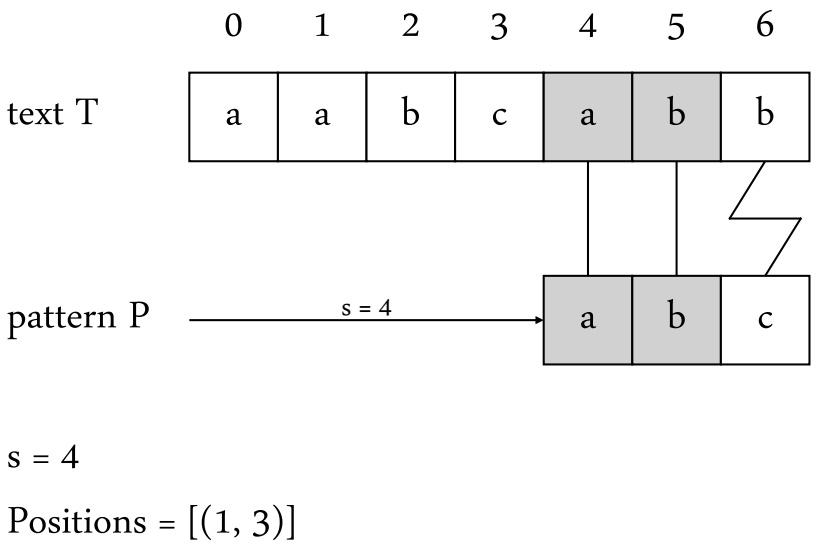



Figure 12: Step 5

- The first two characters in pattern $P$ ('a', 'b') match the corresponding characters in text $T$, but the third character in pattern $P$ ('c') does not match the character in text $T$ ('b') at the current shift position $s = 4$.

  Skip to the next shift without updating *Positions*.

- After that, $s$ increases by 1. Because $s = 5$ and $n - m = 7 - 3 = 4$, we have $s > n - m$. The algorithm terminates and returns the list of found matches: Positions $= [(1, 3)]$.

### 3.2.4 Complexity

| Metric | Complexity | Reasoning |
|---|---|---|
| **Time (Worst)** | $O(K \cdot m \cdot R \cdot C)$ | Occurs when patterns and text share many prefix characters (e.g., all 'A's). |
| **Time (Average)** | $O(K \cdot R \cdot C)$ | In practice, a mismatch usually occurs after only a few character comparisons. |
| **Time (Best)** | $O(K \cdot R \cdot C)$ | The first character of the pattern always results in a mismatch at every shift. |
| **Auxiliary Space** | $O(1)$ | The algorithm uses only a constant amount of extra memory for loop indices. |

Table 3: Complexity Analysis of 2D Brute Force Algorithm

### 3.2.5 Discussion

- In scenarios 1, the number of patterns $K$ is small, and the text is from $10 \times 10$ to $500 \times 500$. Brute-force approach is efficient and likely linear running time $O(R \cdot C)$.

- In scenarios 2, the number of patterns $K$ is large, and the text is fixed at $100 \times 100$. This scenario is more challenging for the brute-force approach, as the running time multiplies by $K$ times, which is much longer compared to scenario 1.

- In conclusion, the brute-force approach is more suitable for scenarios with a small number of patterns. It is worse for scenarios with a large number of one.

## 3.3 The Boyer Moore

### 3.3.1 Introduction & Application

- **Core idea:** The Boyer Moore Algorithm is a practical method to find patterns. This algorithm is the combination of three main ideas comprising scanning the pattern in reverse order, bad character heuristic and good suffix heuristic. This speeds up the searching process considerably by skipping a plenty of characters.
- **Reality applications:**
  - Text Editors: Using in "Search" or "Find and Replace" features to locate the destination of the words.
  - Intrusion Detection Systems (Network Security): use Boyer-Moore to scan network traffic in real-time for "malicious signatures"—specific byte patterns that indicate a cyberattack or virus.
  - Large-Scale Plagiarism Detection: compare submitted essays against billions of web pages and academic journals to find copied content.

### 3.3.2 Pseudocode

**Notation:**

- **Text $T$ ($R$ rows, $C$ columns)**: The 2D character matrix.
- **Pattern $P$ ($K$ keywords)**: The list of strings to search.
- **Heuristics**: $BC$ (Bad Character table) and $GS$ (Good Suffix table).

```
BOYER-MOORE ALGORITHM(T, P, K, R, C)
pos = empty array of size K
for idx = 1 to K do
    P_i = P[idx], m = P_i.length
    BC = PreProcessBadChar(P_i)
    GS = PreProcessGoodSuffix(P_i)
  ▷ Horizontal Search (Rows)
    for r = 0 to R − 1 do
        s = 0
        while s ≤ C − m do
            j = m − 1
            while j ≥ 0 and P_i[j] == T[r, s + j] do
                j = j − 1
            end while
            if j < 0 then
                add (r, s) to pos[idx]
                s = s + GS[0]
            else
                s = s + max(GS[j], j − BC[T[r, s + j]])
```

```
            end if
        end while
    end for
    ▷ Vertical Search (Columns)
    for c = 0 to C − 1 do
        s = 0
        while s ≤ R − m do
            j = m − 1
            while j ≥ 0 and P_i[j] == T[s + j, c] do
                j = j − 1
            end while
            if j < 0 then
                add (s, c) to pos[idx]
                s = s + GS[0]
            else
                s = s + max(GS[j], j − BC[T[s + j, c]])
            end if
        end while
    end for
end for
return pos
```

### 3.3.3 Step-by-step trace

**Notation:**

- **Text** $T$ **(length** $n$**)**: The main text being searched.
- **Pattern** $P$: The string that we are looking for.
- **Shift** ($s$): The current offset of the text T.
- **Positions ([(start position, end position), . . . ])**: A list storing the indices where a complete match is found.

**Example:** text $T = [G, C, A, T, G, C, C, T, A, G]$ and pattern $P = [T, A, T, G]$

- Step 1: Build bad character and good suffix tables base on the pattern we are searching for:

| char | A | T | G |
|---|---|---|---|
| shift value | 1 | 2 | 3 |

Figure 13: Bad character table

| index j | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| shift value | 4 | 4 | 4 | 1 |

Figure 14: Good suffix table

**Note:**

- **Bad character table:** Choose the index for characters from 0 to n - 1. In this case, T appears two times, only the last appearance are regconized.
- **Good suffix table:** For index j from 3 to 0, if the suffix of the character at index j is found, the shift value will be decreased. In this example, index 3 is the exception since there is no suffix and the shift value is 1, other suffixes including g, tg, atg are not found in the prefix of the pattern, then their pattern shifts is the length of the pattern.

- Step 2: Compare the patterns and make shifts based on preprocessing values.

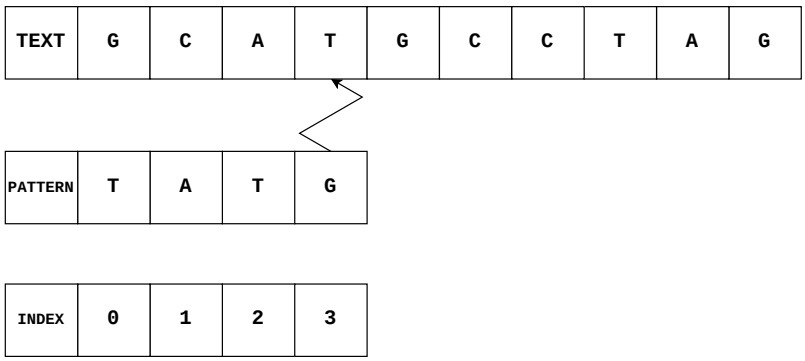



Figure 15: compare pattern from right to left

Mismatch at index j = 3, then calculate the shift value.

shift = max(GS, BC)

GS = 1, BC = max(1, j - lastOccur) = max(1, 3 - 2) = 1

⇒ shift value = 1

- **Continue searching process**

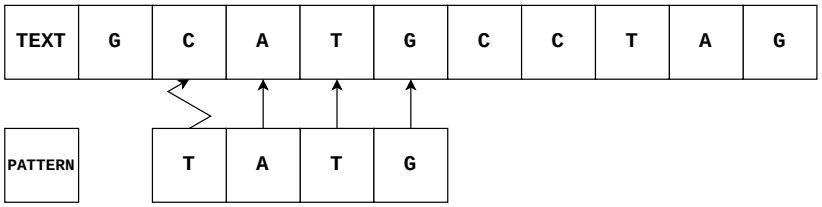



Figure 16: compare pattern

Mismatch at index 0, continue calculate the pattern shift
Shift value = max(GS, BC) = max(4, 1) = 4

- **Shift by 4 characters**

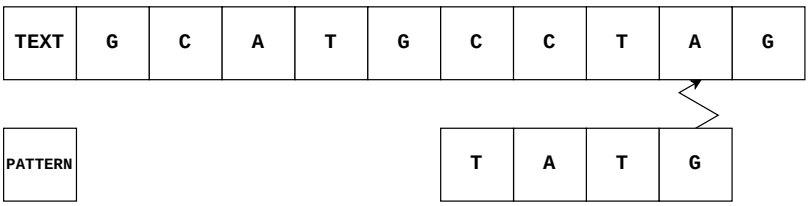



Figure 17: Make larger shift by using good suffix heuristic

Mismatch at index 3, continue calculate shift value
shift value = max(GS, BC) = max(1, 2) = 2

- **Shift by 2 characters**

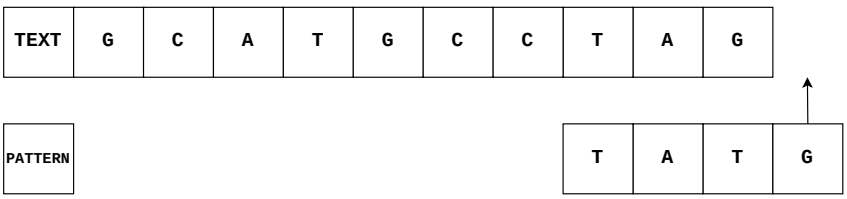



Figure 18: Make bigger shift by using bad character heuristic

The algorithm stops since the pattern is out of text's range.

### 3.3.4 Complexity

| Metric | Complexity | Reasoning |
|---|---|---|
| **Time (Worst)** | $O(K \cdot m \cdot R \cdot C)$ | Occurs when many characters are repeated and both bad character and good suffix rules provide minimal shifts |
| **Time (Average)** | $O(R \cdot C)$ | In reality, the rules of bad character and good suffix usually make larger skips and considerably improve the efficiency of the algorithm. |
| **Time (Best)** | $O(\frac{R \cdot C}{m})$ | In best scenario, mismatches happen at the last character, allowing maximum shifts close to the pattern length(m). |
| **Auxiliary Space** | $O(\lvert\Sigma\rvert + m)$ | The algorithm uses extra memory for bad character table (alphabet size) and good suffix table (pattern length) |

Table 4: Complexity Analysis of 2D Boyer-Moore Algorithm

### 3.3.5 Discussion

- In Scenario 1, where the number of patterns $K$ is small and the text size ranges from $10 \times 10$ to $500 \times 500$, the Boyer–Moore algorithm performs very efficiently due to its ability to skip large portions of the text. The running time is close to $O(R \cdot C)$ in practice.
- In Scenario 2, where the number of patterns $K$ is large and the text is fixed at $100 \times 100$, the total running time increases linearly with $K$. However, Boyer–Moore still outperforms brute-force due to fewer comparisons per pattern.
- In conclusion, the Boyer–Moore algorithm is particularly effective when the pattern is relatively long and the alphabet is large, as it can achieve significant shifts and reduce the number of comparisons.

## 3.4 The Aho-Corasick algorithm

### 3.4.1 Introduction & Application

- **Core idea:** According to Abdeen[1], the brute-force algorithm works with two pointers; a "text pointer" i and a "pattern pointer" j. For all possible shifts of the pattern P over the text T, compare the pattern characters with the corresponding text characters. If mismatch occurs, i is incremented by 1, and j is reset to 0 and the comparing process is restarted. In the case that all characters match, the algorithm records the position of the match and continues searching for the next possible match.
- **Real-world applications:**
  - Text Editors: Using in "Search" or "Find and Replace" features to locate destination words.
  - Bioinformatics: Searching for specific DNA.

- Checking for Plagiarism: Comparing documents to find and compare a script or a piece of code with the exact match in a large database.

### 3.4.2 Pseudocode

**Notation:**

- **Trie Node**: A structure containing a character $pC$, a list of pattern indices $pos$, child pointers $nxt$, transition pointers $go$, a suffix link $link$, and a parent pointer $par$.
- **Text $T$ (R rows, C columns)**: The 2D grid of characters.
- **Pattern $P$ (K keywords)**: The list of patterns to search for.
- **List_string**: A 1D array of size $M = R + C$, containing all rows followed by all columns extracted from $T$.

```
ADD-STRING(S, idx, root)
cur = root
for each character c in S do
    if cur.nxt[c] == null then
        cur.nxt[c] = new Node
        cur.nxt[c].par = cur
        cur.nxt[c].pC = c
    end if
    cur = cur.nxt[c]
end for
append idx to cur.pos
```

```
GET-LINK(u, root)
if u.link == null then
    if u == root or u.par == root then
        u.link = root
    else
        u.link = GO(GET-LINK(u.par), u.pC)
    end if
end if
return u.link
```

```
GO(u, c, root)
if u.go[c] == null then
    if u.nxt[c] ≠ null then
        u.go[c] = u.nxt[c]
    else
        u.go[c] = (u == root) ? u : GO(GET-LINK(u), c)
    end if
end if
return u.go[c]
```

```
SEARCH-STRING(S, IDX, R, P, ans, root)
cur = root
for i = 0 to S.length − 1 do
    cur = GO(cur, S[i])
```

```
    tmp = cur
    while tmp ≠ root do
        for each idx in tmp.pos do
            m = P[idx].length
            if IDX < R then    ▷ Horizontal match (Row extraction)
                start = (IDX, i − m + 1)
                end = (IDX, i)
            else    ▷ Vertical match (Column extraction)
                start = (i − m + 1, IDX − R)
                end = (i, IDX − R)
            end if
            append (start, end) to ans[idx]
        end for
        tmp = GET-LINK(tmp)
    end while
end for
```

```
AHO-CORASICK(T, P, K, R, C)
List_string = CONVERT-GRID-TO-STRINGS(T)
ans = array of K empty lists
root = new Node
maxLen = max(R, C)
    ▷ Build the Trie
for i = 0 to K − 1 do
    if P[i].length ≤ maxLen then
        ADD-STRING(P[i], i, root)
    end if
end for
    ▷ Process all rows and columns
for i = 0 to List_string.length − 1 do
    SEARCH-STRING(List_string[i], i, R, P, ans, root)
end for

REMOVE-TRIE(root)    ▷ Free allocated memory
return ans
```

### 3.4.3 Step-by-step trace[2]

**Notation:**

- **Text** $T$ **(length** $n$**)**: The main text being searched.
- **Pattern** $P$: The string that we are looking for.
- **Shift** ($s$): The current offset of the text T.
- **Positions ([(start position, end position), . . . ])**: A list storing the indices where a complete match is found.

**Example:** text $T = [a, a, b, c, a, b, b]$ and pattern $P = [a, b, c]$

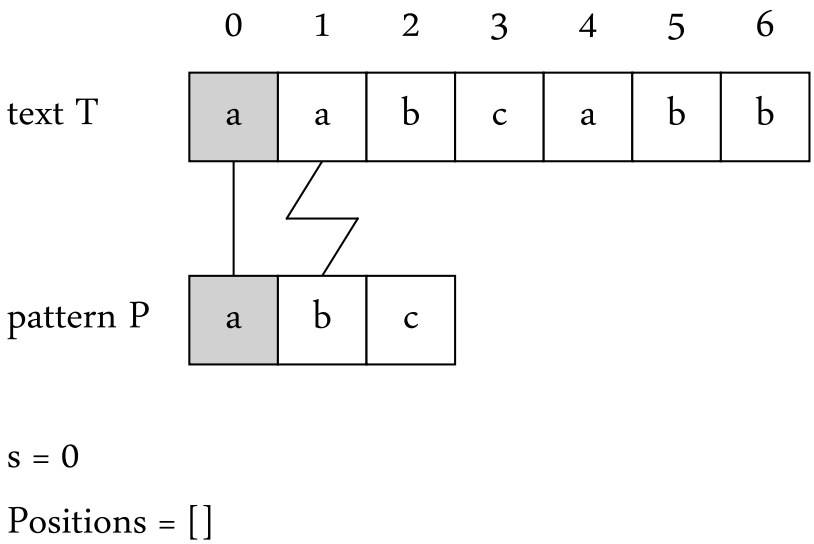



Figure 19: Step 1

- The first character 'a' matches in both arrays.

  The second character in text $T$ ('a') does not match the second character in pattern $P$ ('b').

  Text at the bottom shows the current shift is $s = 0$ and the list of found matches is empty ($Positions = []$).

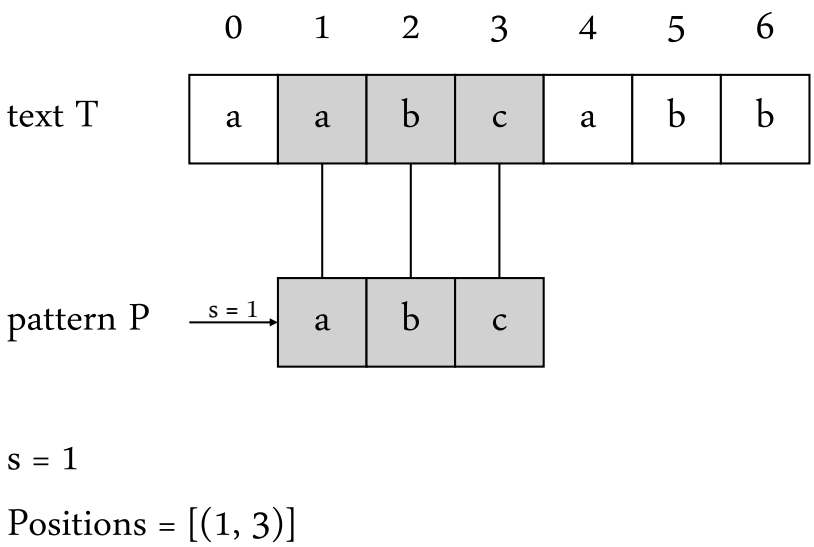
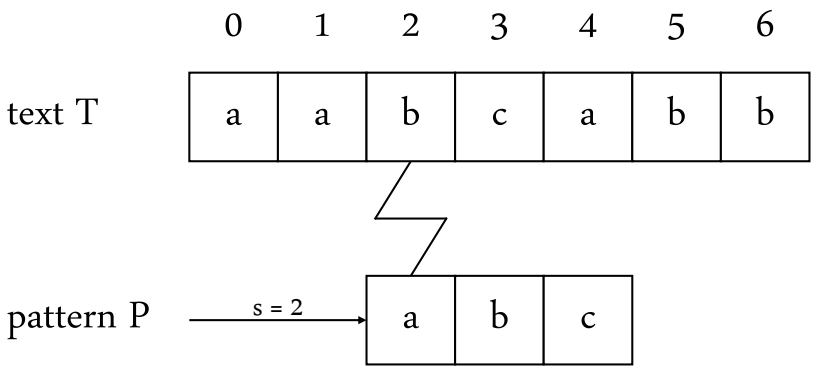



Figure 20: Step 2

- All characters in pattern $P$ ('a', 'b', 'c') successfully match the corresponding characters in text $T$ ('a', 'b', 'c').

  Shift is $s = 1$ and the list of found matches is updated to record this position ($Positions = [(1, 3)]$).



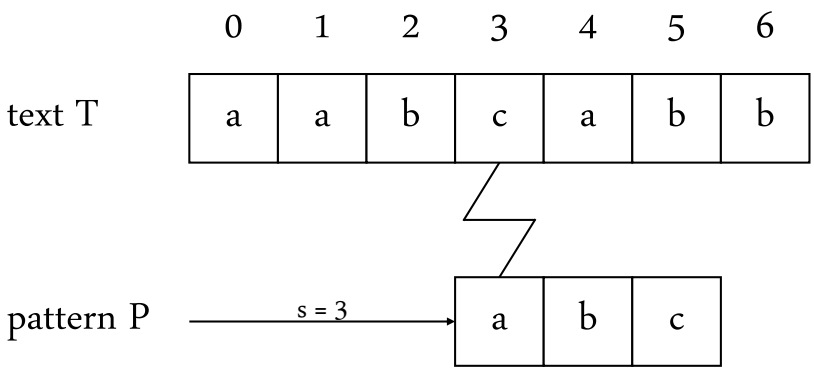

s = 2

Positions = [(1, 3)]

Figure 21: Step 3

- The first character in pattern $P$ ('a') does not match the character in text $T$ ('b') at the current shift position $s = 2$.

  Skip to the next shift without updating *Positions*.




s = 3

Positions = [(1, 3)]

Figure 22: Step 4

- The first character in pattern $P$ ('a') does not match the character in text $T$ ('c') at the current shift position $s = 2$.

  Skip to the next shift without updating *Positions*.

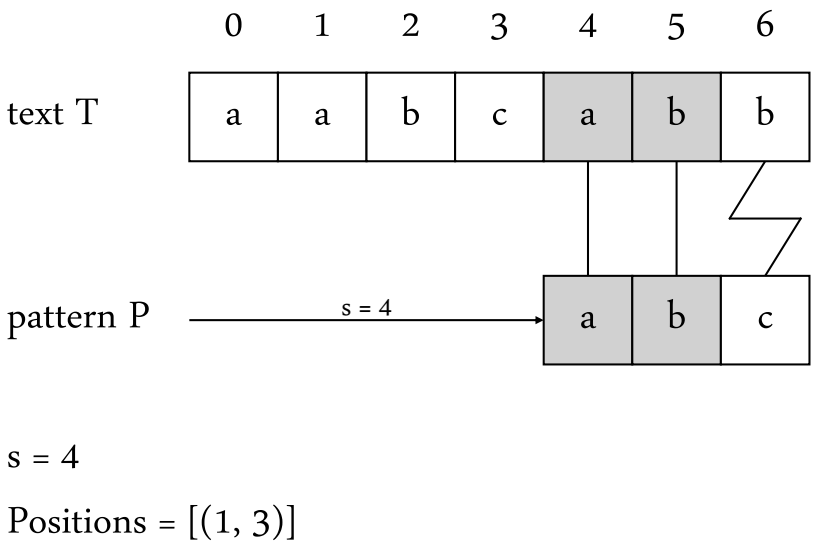



Figure 23: Step 5

- The first two characters in pattern $P$ ('a', 'b') match the corresponding characters in text $T$, but the third character in pattern $P$ ('c') does not match the character in text $T$ ('b') at the current shift position $s = 4$.

  Skip to the next shift without updating *Positions*.

- After that, $s$ increases by 1. Because $s = 5$ and $n - m = 7 - 3 = 4$, we have $s > n - m$. The algorithm terminates and returns the list of found matches: Positions $= [(1, 3)]$.

### 3.4.4 Complexity

| Metric | Complexity | Reasoning |
|---|---|---|
| **Time (Worst)** | $O(K \cdot m \cdot R \cdot C)$ | Occurs when patterns and text share many prefix characters (e.g., all 'A's). |
| **Time (Average)** | $O(K \cdot R \cdot C)$ | In practice, a mismatch usually occurs after only a few character comparisons. |
| **Time (Best)** | $O(K \cdot R \cdot C)$ | The first character of the pattern always results in a mismatch at every shift. |
| **Auxiliary Space** | $O(1)$ | The algorithm uses only a constant amount of extra memory for loop indices. |

Table 5: Complexity Analysis of 2D Brute Force Algorithm

### 3.4.5 Discussion

- In scenarios 1, the number of patterns $K$ is small, and the text is from $10 \times 10$ to $500 \times 500$. Brute-force approach is efficient and likely linear running time $O(R \cdot C)$.

- In scenarios 2, the number of patterns $K$ is large, and the text is fixed at $100 \times 100$. This scenario is more challenging for the brute-force approach, as the running time multiplies by $K$ times, which is much longer compared to scenario 1.

- In conclusion, the brute-force approach is more suitable for scenarios with a small number of patterns. It is worse for scenarios with a large number of one.

# 4 Experimental Results

## 4.1 Running time

The four line graphs below illustrate the running times of 12 sorting algorithms across four input data distributions: **sorted, nearly sorted, reverse sorted, and random**.

Overall, it is evident that **Selection Sort** and **Bubble Sort** exhibit significantly higher execution times compared to the other algorithms. Notably, **Counting Sort** remains the most efficient algorithm across all test cases.

In the sorted and nearly sorted scenarios, **Selection Sort** and **Bubble Sort** consistently record the highest running times.

Regarding the reverse sorted and randomized input, five algorithms show a dramatic increase in execution time: **Insertion Sort**, **Binary Insertion Sort**, **Shaker Sort**, and the two previously mentioned algorithms (i.e., **Selection Sort** and **Bubble Sort**).

## 4.2 The number of comparisons

The four bar charts below display the number of comparisons made by 12 different categories of sorting algorithms across four input data types: **sorted, nearly sorted, reverse sorted, and random**.

Generally, it is clear that **Selection Sort** and **Bubble Sort** consistently require the highest number of comparisons across all scenarios. In constrast, **Counting Sort** exibits the lowest qunatity of comparison count.

In the sorted and nearly sorted cases, **Selection Sort** and **Bubble Sort** perform the remarkably badly, with their comparion counts being significantly higher than the other algorithms.

In the reverse sorted and randomized input scenarios, the number of comparisons for most $O(n^2)$ algorithms (specifically **Insertion Sort**, **Binary Insertion Sort**, and **Shaker Sort**) increases dramatically. Meanwhile, $O(n \log n)$ algorithms such as **Merge Sort**, **Heap Sort**, and **Quick Sort**, etc. maintain a relatively stable and efficient number of comparisons.

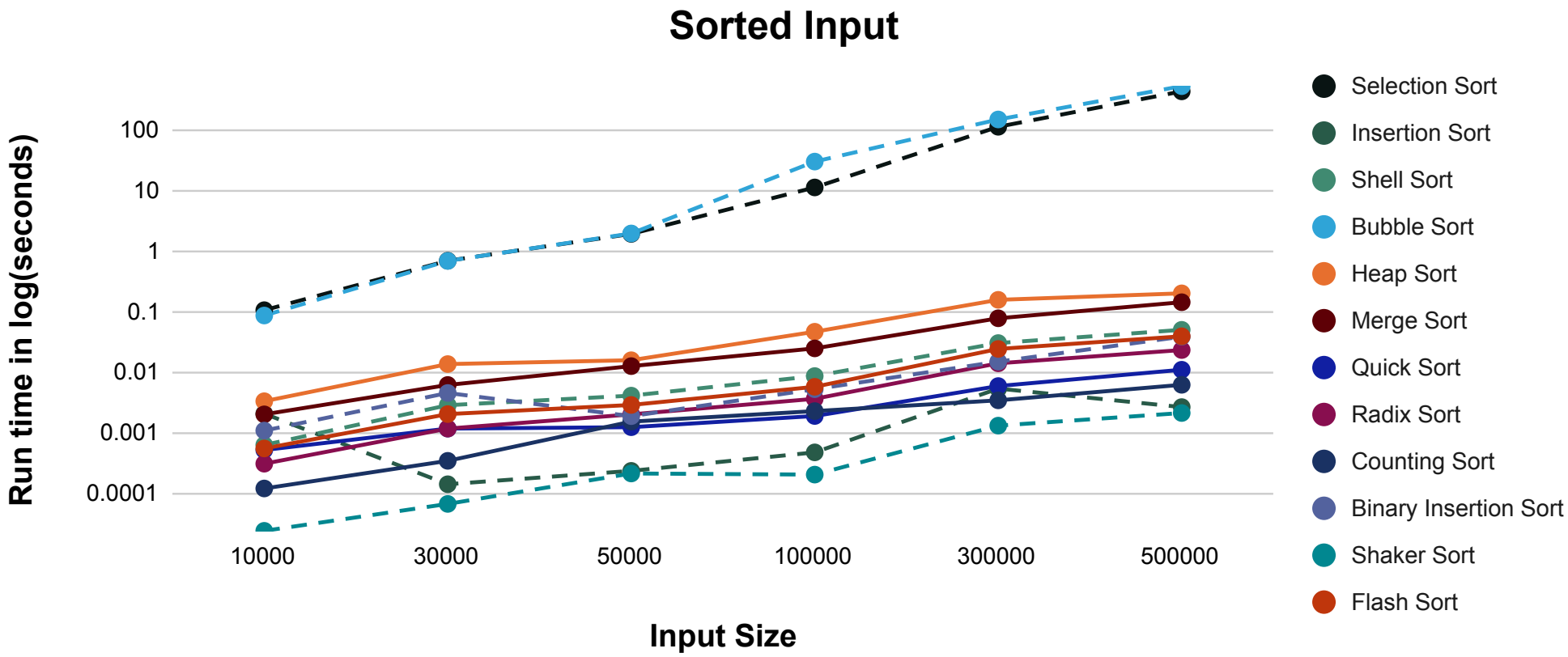
## 4.3 Tables and Graphs

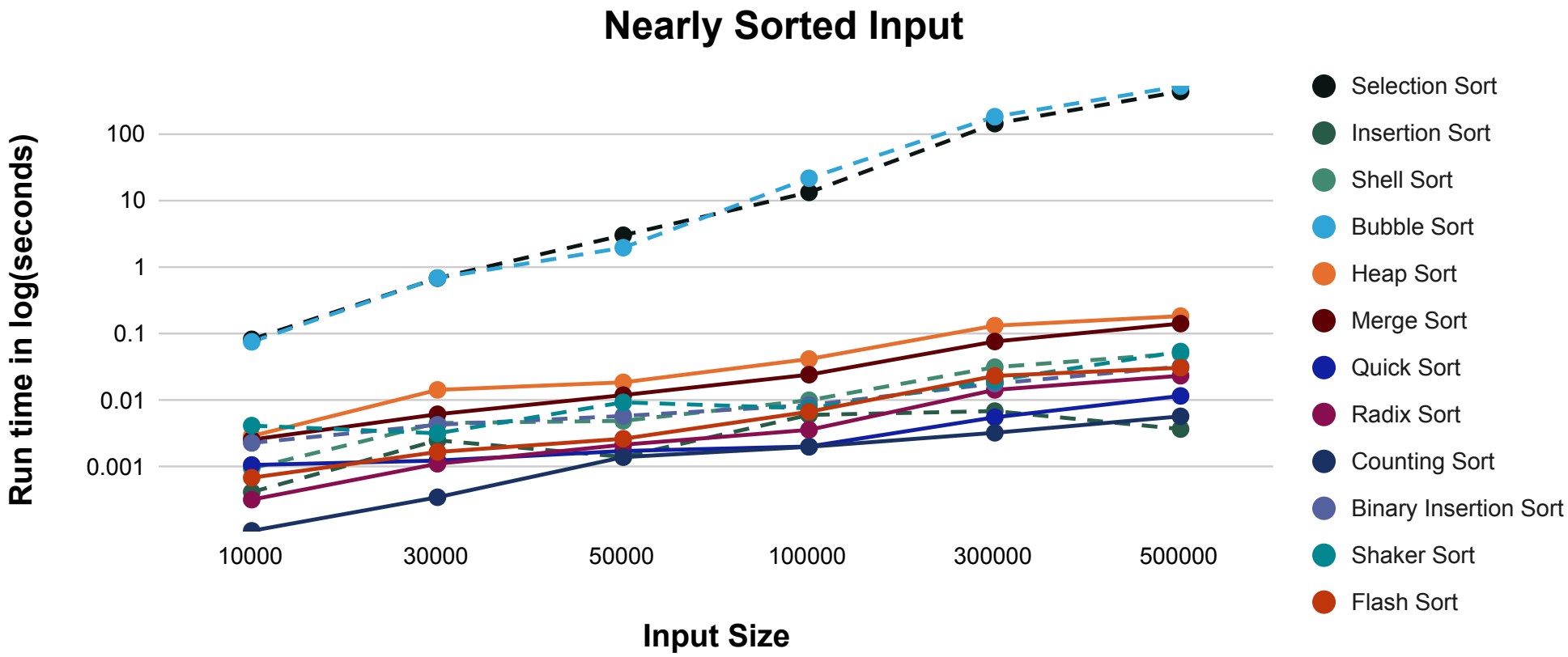
| Data order: **Sorted** | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Data size** | 10.000 | | 30.000 | | 50.000 | | 100.000 | | 300.000 | | 500.000 | |
| **Resulting statics** | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision |
| Selection Sort | 105,167 | 100009999 | 689,787 | 900029999 | 1917,92 | 2500049999 | 11233,5 | 10000099999 | 112996 | 90000299999 | 434892 | 2,5E+11 |
| Insertion Sort | 2,0494 | 29998 | 0,1396 | 89998 | 0,2322 | 149998 | 0,4694 | 299998 | 5,3183 | 899998 | 2,6309 | 1499998 |
| Shell Sort | 0,6222 | 360042 | 2,8272 | 1170050 | 4,0681 | 2100049 | 8,5708 | 4500051 | 30,0859 | 15300061 | 49,7864 | 25500058 |
| Bubble Sort | 85,8152 | 100009999 | 685,405 | 900029999 | 1949,68 | 2500049999 | 30057,4 | 10000099999 | 149342 | 90000299999 | 530168 | 2,5E+11 |
| Heap Sort | 3,289 | 518705 | 13,5221 | 1739633 | 15,6552 | 3056481 | 46,292 | 6519813 | 155,882 | 21431637 | 199,605 | 37116275 |
| Merge Sort | 1,9953 | 465243 | 6,1066 | 3534740 | 12,507 | 11271929 | 24,5689 | 25699104 | 77,2574 | 62516397 | 142,383 | 153959690 |
| Quick Sort | 0,5135 | 244975 | 1,1606 | 823646 | 1,2239 | 1467264 | 1,8735 | 3134498 | 5,8564 | 10258249 | 10,8987 | 17737894 |
| Radix Sort | 0,3049 | 60107 | 1,1599 | 210131 | 1,9931 | 350131 | 3,6059 | 700131 | 13,9273 | 2400155 | 23,0738 | 4000155 |
| Counting Sort | 0,1188 | 70005 | 0,3401 | 210005 | 1,513 | 350005 | 2,2584 | 700005 | 3,4015 | 2100005 | 6,1422 | 3500005 |
| Binary Insertion Sort | 1,076 | 277232 | 4,501 | 924464 | 1,8637 | 1618928 | 5,1965 | 3437856 | 14,8291 | 11251424 | 38,3446 | 19451424 |
| Shaker Sort | 0,0237 | 20001 | 0,066 | 60001 | 0,2102 | 100001 | 0,2014 | 200001 | 1,2986 | 600001 | 2,0998 | 1000001 |
| Flash Sort | 0,5451 | 118994 | 2,0147 | 356994 | 2,8529 | 594994 | 5,6932 | 1189994 | 24,029 | 3569994 | 38,9354 | 5949994 |

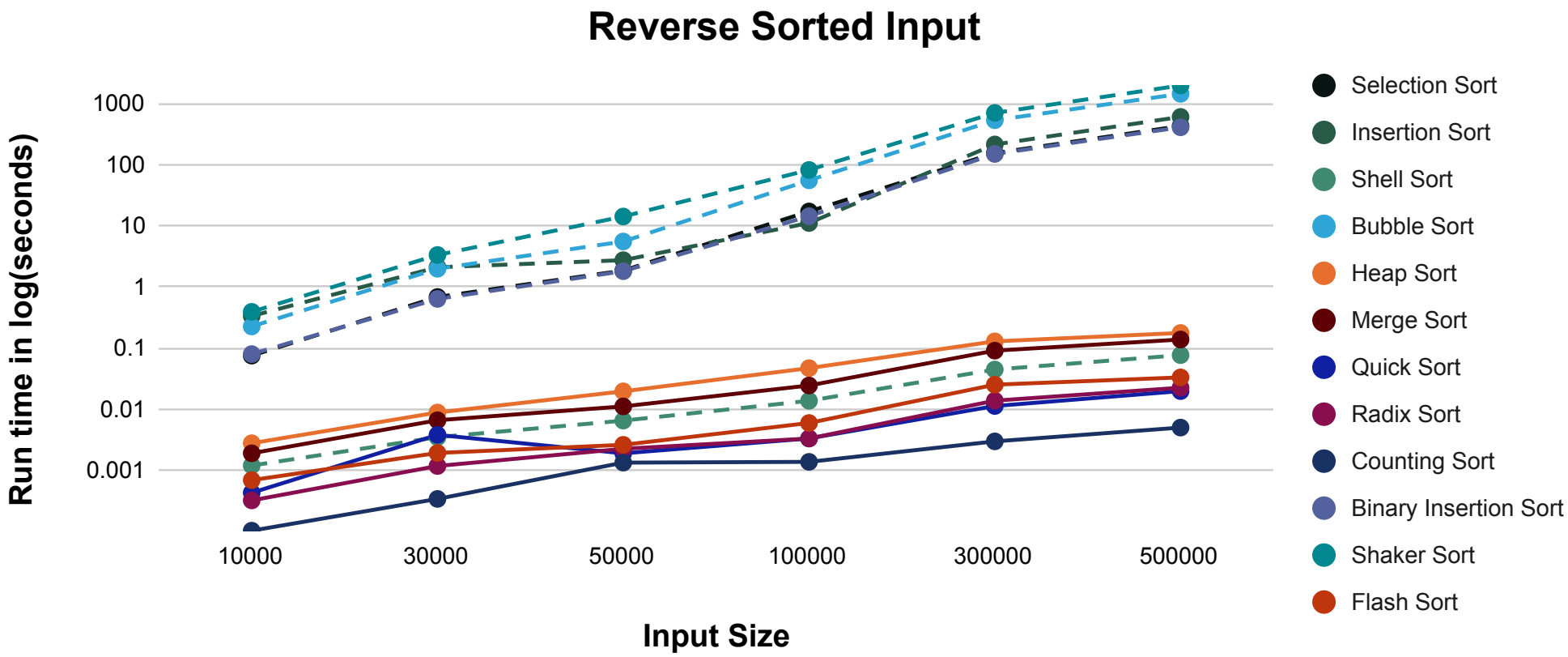
| Data order: **Nearly Sorted** | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Data size** | 10.000 | | 30.000 | | 50.000 | | 100.000 | | 300.000 | | 500.000 | |
| **Resulting statics** | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision |
| Selection Sort | 82,4528 | 100009999 | 682,385 | 900029999 | 3006,82 | 2500049999 | 13195,6 | 10000099999 | 141754 | 90000299999 | 429977 | 2,5E+11 |
| Insertion Sort | 0,4216 | 174730 | 2,5244 | 485886 | 1,4237 | 597658 | 6,0644 | 877238 | 6,9249 | 1442810 | 3,7205 | 2032738 |
| Shell Sort | 0,9578 | 360042 | 4,5803 | 1170050 | 4,9382 | 2100049 | 10,0758 | 4500051 | 31,5097 | 15300061 | 50,6101 | 25500058 |
| Bubble Sort | 75,6924 | 100009999 | 686,235 | 900029999 | 1953,2 | 2500049999 | 21510,8 | 10000099999 | 180087 | 90000299999 | 516635 | 2,5E+11 |
| Heap Sort | 2,9228 | 518705 | 14,3961 | 1739633 | 18,676 | 3056481 | 41,7754 | 6519813 | 132,22 | 21431637 | 183,995 | 37116275 |
| Merge Sort | 2,5839 | 964668 | 6,2076 | 5147796 | 12,0211 | 14036437 | 24,3352 | 31436444 | 76,5169 | 80950733 | 142,079 | 185575804 |
| Quick Sort | 1,077 | 257954 | 1,2535 | 855622 | 1,7467 | 1507515 | 2,0375 | 3184601 | 5,581 | 10309309 | 11,6933 | 17786083 |
| Radix Sort | 0,3253 | 60107 | 1,1225 | 210131 | 2,1678 | 350131 | 3,622 | 700131 | 14,3872 | 2400155 | 23,4548 | 4000155 |
| Counting Sort | 0,1108 | 70005 | 0,3532 | 210005 | 1,4122 | 350005 | 2,0138 | 700005 | 3,2722 | 2100005 | 5,7755 | 3500005 |
| Binary Insertion Sort | 2,3063 | 349914 | 4,3183 | 1175428 | 5,8485 | 1794400 | 8,5103 | 3666698 | 17,885 | 11462644 | 32,2496 | 19594008 |
| Shaker Sort | 4,192 | 260001 | 3,2138 | 780001 | 9,3725 | 1700001 | 7,6127 | 3400001 | 19,9086 | 9000001 | 54,0995 | 15000001 |
| Flash Sort | 0,694 | 118970 | 1,6874 | 356968 | 2,6621 | 594968 | 6,765 | 1189970 | 23,3154 | 3569970 | 30,9164 | 5949971 |

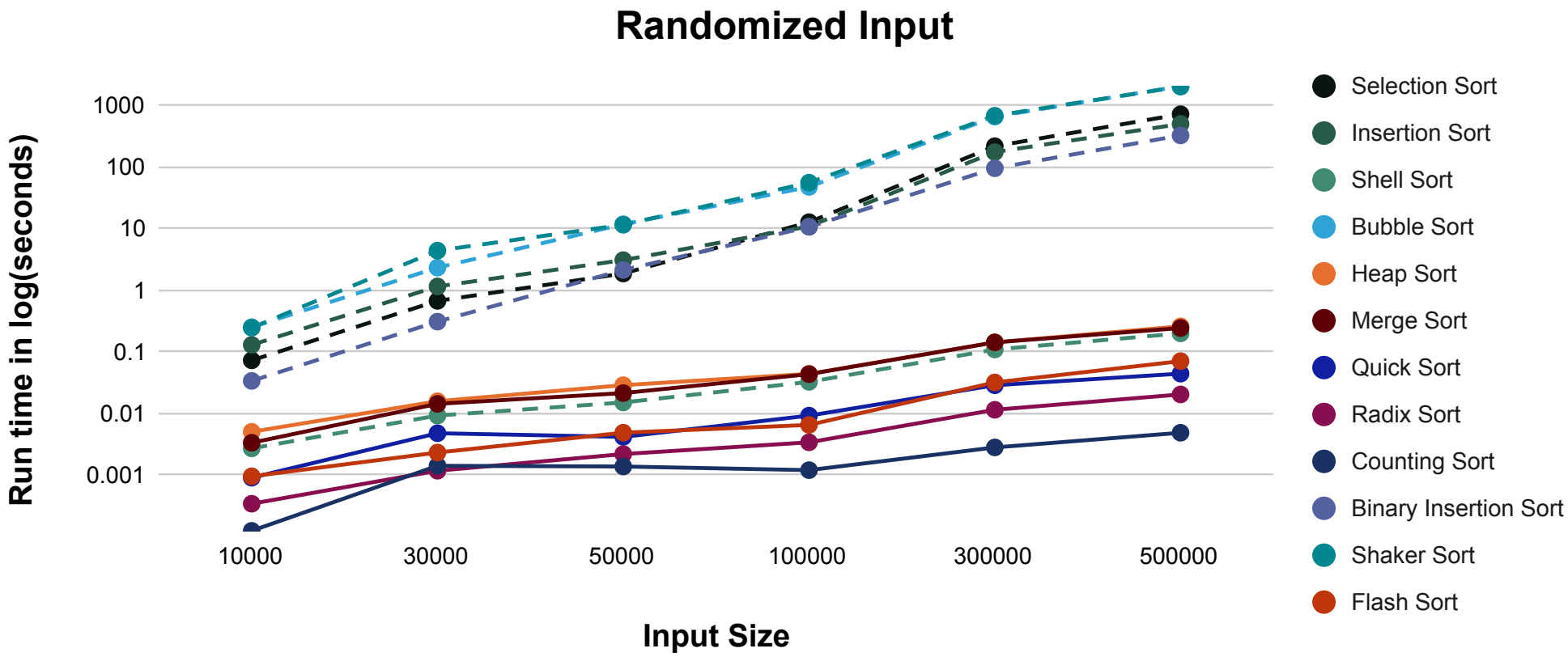
| Data order: **Reverse Sorted** | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Data size** | 10.000 | | 30.000 | | 50.000 | | 100.000 | | 300.000 | | 500.000 | |
| **Resulting statics** | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision |
| Selection Sort | 74,7506 | 100009999 | 680,337 | 900029999 | 1838,97 | 2500049999 | 17064,2 | 10000099999 | 153636 | 90000299999 | 425904 | 2,5E+11 |
| Insertion Sort | 332,46 | 100009999 | 2085,1 | 900029999 | 2722,32 | 2500049999 | 11039,5 | 10000099999 | 212837 | 90000299999 | 6,01E+05 | 2,5E+11 |
| Shell Sort | 1,1906 | 360042 | 3,4599 | 1170050 | 6,4618 | 2100049 | 13,6358 | 4500051 | 44,2755 | 15300061 | 76,1319 | 25500058 |
| Bubble Sort | 224,127 | 100009999 | 1968,73 | 900029999 | 5503,08 | 2500049999 | 55076,2 | 10000099999 | 530188 | 90000299999 | 1431660 | 2,5E+11 |
| Heap Sort | 2,7592 | 518705 | 8,7699 | 1739633 | 19,4824 | 3056481 | 46,6272 | 6519813 | 127,902 | 21431637 | 175,502 | 37116275 |
| Merge Sort | 1,8889 | 1431110 | 6,5497 | 6691262 | 11,0356 | 16720383 | 24,2663 | 37104342 | 90,0728 | 99359047 | 137,091 | 217412214 |
| Quick Sort | 0,4261 | 254764 | 3,779 | 855622 | 1,8849 | 1508351 | 3,3012 | 3219401 | 11,198 | 10567016 | 19,6311 | 18332891 |
| Radix Sort | 0,3199 | 60107 | 1,1688 | 210131 | 2,2355 | 350131 | 3,2913 | 700131 | 13,7003 | 2400155 | 22,3289 | 4000155 |
| Counting Sort | 0,1023 | 70005 | 0,3383 | 210005 | 1,3279 | 350005 | 1,3663 | 700005 | 2,9557 | 2100005 | 4,9697 | 3500005 |
| Binary Insertion Sort | 79,1308 | 50252260 | 632,637 | 450849494 | 1797,31 | 1251493960 | 14267,9 | 5003187890 | 148769 | 45010501462 | 406047 | 1,2502E+11 |
| Shaker Sort | 389,034 | 200000001 | 3311,6 | 1800000001 | 14107 | 5000000001 | 81458,4 | 20000000001 | 697884 | 1,8E+11 | 1963510 | 5E+11 |
| Flash Sort | 0,6865 | 103752 | 1,9107 | 311252 | 2,5955 | 518752 | 5,9086 | 1037502 | 25,004 | 3112502 | 32,9099 | 5187502 |

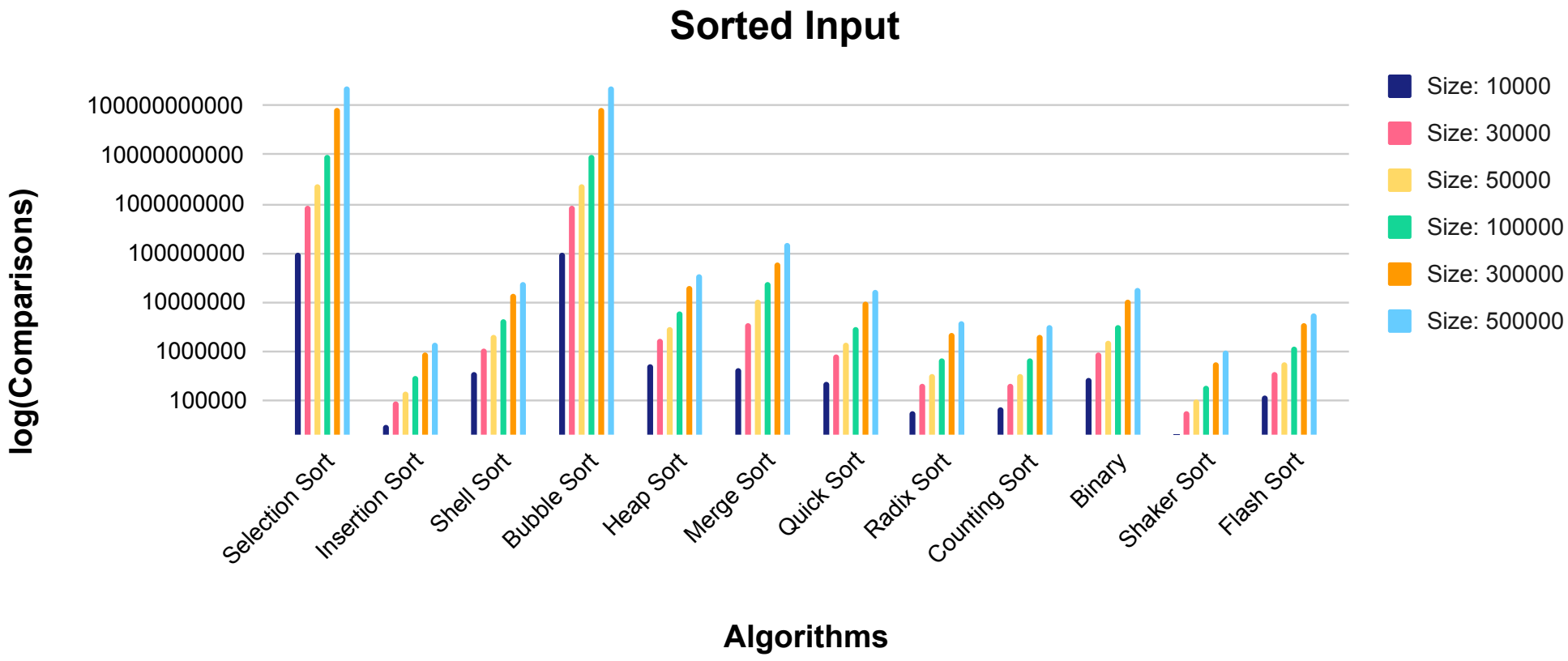
| Data order: **Randomized** | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Data size** | 10.000 | | 30.000 | | 50.000 | | 100.000 | | 300.000 | | 500.000 | |
| **Resulting statics** | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision | Running time | Comparision |
| Selection Sort | 72,8876 | 100009999 | 675,09 | 900029999 | 1878,82 | 2500049999 | 12825,3 | 10000099999 | 220180 | 90000299999 | 728805 | 2,5E+11 |
| Insertion Sort | 130,055 | 49727376 | 1158,94 | 453956164 | 3102,37 | 1249929123 | 10960,5 | 4990211473 | 176560 | 45034384243 | 506017 | 1,251E+11 |
| Shell Sort | 2,7062 | 360042 | 9,2453 | 1170050 | 15,0898 | 2100049 | 32,5809 | 4500051 | 109,411 | 15300061 | 200,162 | 25500058 |
| Bubble Sort | 252,128 | 100009999 | 2342,79 | 900029999 | 11864,4 | 2500049999 | 47993,9 | 10000099999 | 666147 | 90000299999 | 2081340 | 2,5E+11 |
| Heap Sort | 5,033 | 517875 | 15,7985 | 1740722 | 28,8383 | 3059571 | 43,9528 | 6521229 | 142,695 | 21444835 | 260,311 | 36875028 |
| Merge Sort | 3,3413 | 2004825 | 14,3255 | 8599102 | 21,4229 | 20053445 | 43,5151 | 44170450 | 142,988 | 122441839 | 244,4 | 257294351 |
| Quick Sort | 0,9036 | 314867 | 4,7699 | 1126187 | 4,1647 | 2019129 | 9,22 | 4395007 | 28,7321 | 14774390 | 44,3427 | 28182172 |
| Radix Sort | 0,3412 | 60107 | 1,1721 | 210131 | 2,1878 | 350131 | 3,3941 | 700131 | 11,4715 | 2400155 | 20,378 | 4000155 |
| Counting Sort | 0,1238 | 69999 | 1,4104 | 210005 | 1,3746 | 315541 | 1,1992 | 565541 | 2,805 | 1565541 | 4,857 | 2565541 |
| Binary Insertion Sort | 33,8242 | 25162414 | 310,545 | 225998727 | 2143,52 | 622988577 | 10754,9 | 2506869309 | 96353,5 | 22539532279 | 329530 | 6,2562E+10 |
| Shaker Sort | 247,509 | 101180001 | 4425,13 | 895440001 | 11648,6 | 2499700001 | 56346,5 | 10061400001 | 688766 | 90054600001 | 2048100 | 2,4994E+11 |
| Flash Sort | 0,9527 | 96409 | 2,3163 | 288519 | 4,8677 | 466016 | 6,5136 | 881806 | 31,9216 | 2550156 | 70,7807 | 4543029 |

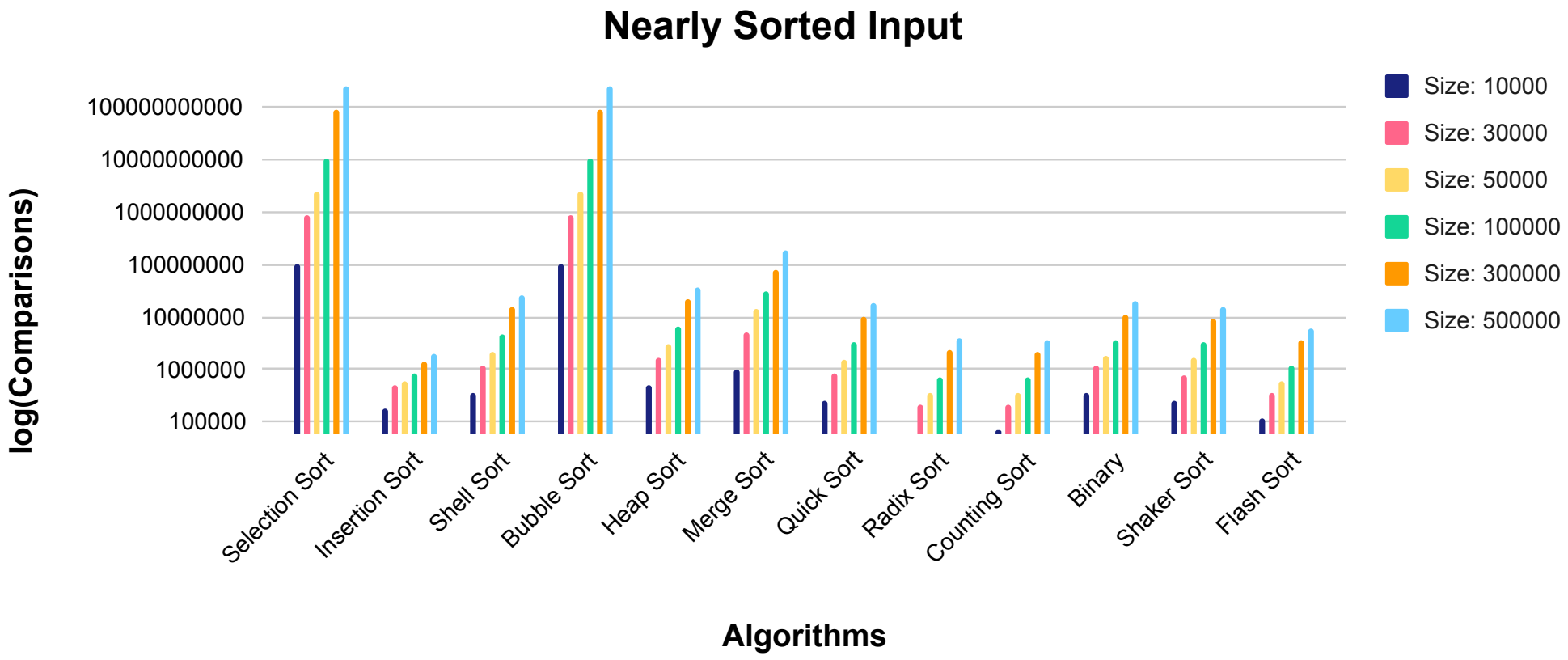

Sorted Input
Run time in log(seconds)
100
10
1
0.1
0.01
0.001
0.0001
10000
30000
50000
100000
300000
500000
Input Size
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary Insertion Sort
Shaker Sort
Flash Sort

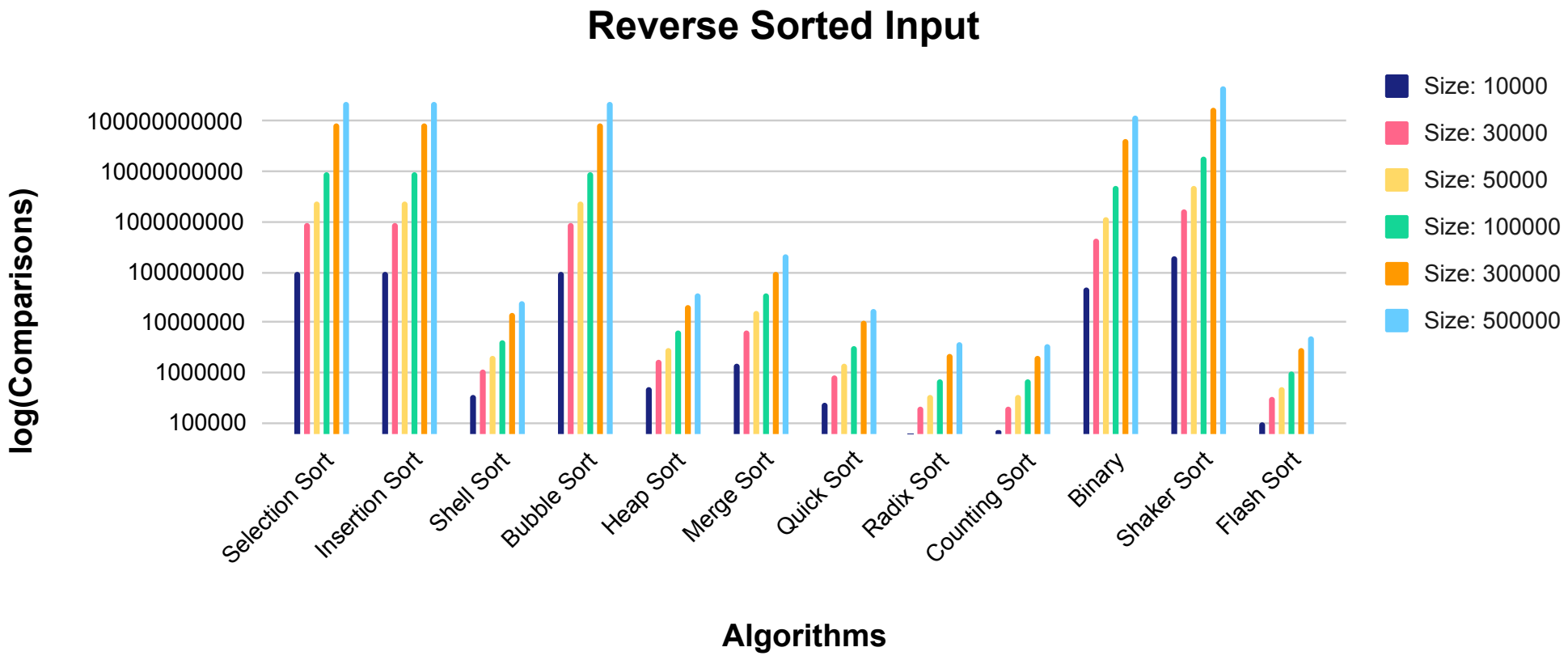

Nearly Sorted Input
Run time in log(seconds)
100
10
1
0.1
0.01
0.001
10000
30000
50000
100000
300000
500000
Input Size
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary Insertion Sort
Shaker Sort
Flash Sort

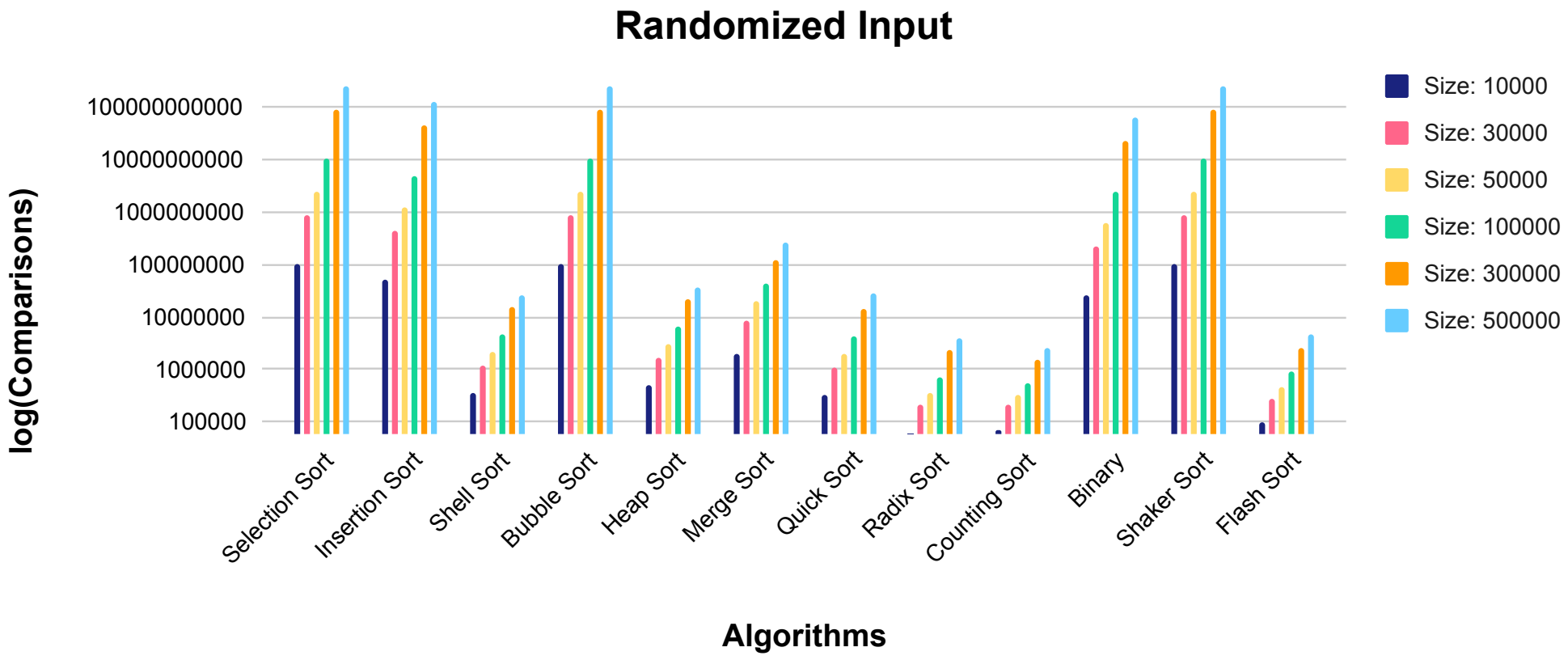

Reverse Sorted Input
Run time in log(seconds)
1000
100
10
1
0.1
0.01
0.001
10000
30000
50000
100000
300000
500000
Input Size
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary Insertion Sort
Shaker Sort
Flash Sort


Randomized Input
Run time in log(seconds)
1000
100
10
1
0.1
0.01
0.001
10000
30000
50000
100000
300000
500000
Input Size
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary Insertion Sort
Shaker Sort
Flash Sort


Sorted Input
log(Comparisons)
100000000000
10000000000
1000000000
100000000
10000000
1000000
100000
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary
Shaker Sort
Flash Sort
Algorithms
Size: 10000
Size: 30000
Size: 50000
Size: 100000
Size: 300000
Size: 500000


Nearly Sorted Input
log(Comparisons)
100000000000
10000000000
1000000000
100000000
10000000
1000000
100000
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary
Shaker Sort
Flash Sort
Algorithms
Size: 10000
Size: 30000
Size: 50000
Size: 100000
Size: 300000
Size: 500000


Reverse Sorted Input
log(Comparisons)
100000000000
10000000000
1000000000
100000000
10000000
1000000
100000
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary
Shaker Sort
Flash Sort
Algorithms
Size: 10000
Size: 30000
Size: 50000
Size: 100000
Size: 300000
Size: 500000


Randomized Input
log(Comparisons)
100000000000
10000000000
1000000000
100000000
10000000
1000000
100000
Selection Sort
Insertion Sort
Shell Sort
Bubble Sort
Heap Sort
Merge Sort
Quick Sort
Radix Sort
Counting Sort
Binary
Shaker Sort
Flash Sort
Algorithms
Size: 10000
Size: 30000
Size: 50000
Size: 100000
Size: 300000
Size: 500000

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# 6 References

## References

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